

Structure-Based UAV Visual Geo-Localization via Delaunay Triangulation and Ekeland Free-Cone Angle Descriptors

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Abstract

Accurate geo-localization of unmanned aerial vehicles (UAVs) in GPS-denied environments remains a core challenge for autonomous navigation. We study a scene-agnostic, training-free approach that seeks the UAV’s 2-D geographic position by matching purely geometric building-footprint descriptors extracted from a single nadir image against a geo-referenced satellite map, and we provide a rigorous analysis of where and why this appealing paradigm succeeds and fails. Buildings are segmented by a pre-trained Mask R-CNN; each polygon is decomposed into triangles by Constrained Delaunay Triangulation (CDT), and each triangle is encoded with a 5-dimensional descriptor $f = (\alpha_1, \alpha_2, e_1, e_2, e_3)$ comprising two sorted interior angles and three **Maximal Free Cone Angles (MFCA)**—a geometric quantity measuring the largest obstacle-free angular sector at each vertex. The MFCA is directly inspired by Ekeland’s variational principle (1974): the principle guarantees a critical point where no improvement is achievable without a proportional distance cost; we instantiate this as a *kernel expansion* P_{sub} —the largest polygon formed by merging adjacent CDT triangles such that all three original triangle vertices remain on the boundary of the result. This stopping criterion is the direct geometric analogue of Ekeland’s cone-criticality condition: the polygon cannot be expanded further while keeping the triangle vertices as boundary (critical) points. The Ekeland angle e_v at each vertex is then the exterior angle of P_{sub} at v , computed in $O(1)$ per vertex by an edge-anchored half-plane construction with no sweep required. Satellite patch descriptors are indexed offline in a KD-tree (ℓ_1 metric); online matching runs in $O(M \log N)$ time per query. We evaluate this pipeline rigorously on the UAV-VisLoc benchmark and report a central negative result with a precise diagnosis. Building coverage at the true location is high (93%), yet single-frame descriptor retrieval fails at city scale: the geometrically correct match is essentially never among the top-ranked candidates (retrieval $<1\%$) and localization error remains at the kilometre level. This ceiling is one of descriptor *discriminability*, not of coverage or feature scaling—enriching the descriptor with building size and a log-polar building-constellation context, matching at building rather than triangle granularity, and normalizing feature dimensions each shrink the true-match descriptor distance by up to $16\times$ yet leave retrieval near zero. Following the coarse-to-fine, particle-filter paradigm of successful vector-map methods, we further integrate the per-frame matches over a flight trajectory with a Gaussian-mixture observation and a particle filter; this too fails to converge, because the true position receives recurring observation support in only 12% of frames. This limit is terrain-general: across five UAV-VisLoc sites spanning urban, coastal, hilly, built-up, and arid terrain, meter-level accuracy stays below 3% within 500m with kilometre-scale median error. We conclude that the limiting factor is the *observation model*: purely geometric per-element matching does not supply the recurring, spatially consistent support that sequential estimation requires at scale, and must be paired with a stronger area-level observation. Acting on this diagnosis, we show the fix is constructive and keeps the Ekeland angle central: aggregating the descriptor over ≈ 200 m cells—as a histogram of per-building Ekeland free-cone angles alongside building density and layout—escapes the per-element ceiling and, on every site, improves coarse-localization recall over a density/layout-only descriptor

of the kind CFBVM-style area matching uses (e.g. top-50 recall $2.0\% \rightarrow 10.2\%$ on the coastal site). The Ekeland free-cone angle is thus a complementary, discriminative shape cue at the area level, though absolute accuracy remains coarse. Our analysis provides the missing empirical account of why training-free geometric matching, though attractive, does not by itself scale—and where the Ekeland descriptor nonetheless adds measurable value.

1 Introduction

UAVs depend on GNSS for position fixes in nearly every operational domain: search-and-rescue, precision agriculture, infrastructure inspection, and military reconnaissance. Yet GNSS is unavailable in urban canyons, jammed in contested airspace, and unreliable indoors—scenarios that are precisely those requiring the most reliable autonomy. A single GNSS outage during a search-and-rescue operation can redirect a UAV kilometres from the target zone. In contested airspace, GPS jamming has been documented to deny localization entirely for extended periods [11, 25]. For emergency responders, infrastructure inspectors, and autonomous delivery operators, GNSS failure is not a theoretical edge case—it is the operational norm in precisely the environments these platforms are deployed. Vision-based absolute visual localization (AVL) addresses this gap by recovering the UAV’s 2-D geographic position from onboard camera images matched against a pre-stored reference map, with no satellite signal required [6, 28].

The dominant difficulty in UAV-to-satellite AVL is the *cross-view appearance gap*: satellite maps are orthorectified nadir projections captured in a fixed season, while UAV images arrive at varying altitude, heading, roll, pitch, time of day, and season. Existing methods expose a fundamental trilemma: (i) *Appearance-based methods* (SIFT, ORB, deep cross-view networks [13, 15]) rely on pixel-level signals that change across season, altitude, and lighting—SIFT achieves zero Recall@10 on MAFS-10 after 1,630 s of computation [34], confirming that texture descriptors cannot bridge the nadir gap. (ii) *Deep learning methods* overcome the appearance gap by learning invariant embeddings, but require scene-specific retraining, making them impossible to deploy on a new city map within hours. (iii) *Vector map methods* exploit shape rather than texture, but existing approaches (BRM [4], CFBVM [28]) operate on whole-building statistics and lose the *spatial context between buildings*, making them poorly discriminative in dense urban blocks where many buildings share similar individual shapes. No existing training-free method simultaneously encodes both triangle-level shape and the structural context of the surrounding building footprint within a single descriptor.

Our key insight is that *building footprint geometry* is stable across all of these appearance variations: a polygon extracted from a nadir UAV image and the same polygon in a satellite map agree in shape regardless of altitude, season, or illumination. We exploit this by (i) decomposing each building polygon into triangles via CDT and (ii) encoding each triangle with a 5-D descriptor of pure angular quantities invariant to scale, rotation, and appearance. The free-space component of this descriptor—the Maximal Free Cone Angle (MFCA)—is directly inspired by Ekeland’s variational principle [12]: the principle characterises a point of maximal local stability as the apex of the largest cone that touches but does not cross a function’s graph; we instantiate this cone not over a function graph but over a 2-D polygon boundary, yielding a scalar that encodes *how much local geometric freedom exists at each building vertex*.

Training-free geometric localization is an appealing route to *map-portable autonomy*—a platform pre-loaded with satellite imagery of any new city operating immediately, with no training phase. This paper asks, and answers rigorously, whether purely geometric per-element matching can deliver it: we find that it cannot at city scale, and we characterize precisely why, turning an aspirational claim into a measured account of the paradigm’s limits and of what a viable observation model must provide.

Contributions.

1. **Edge-anchored Ekeland angle (MFCA).** We derive the Ekeland angle e_v from first principles via a two-phase construction: (i) expand the kernel triangle into a maximal polygon P_{sub} whose stopping condition is geometrically analogous to Ekeland’s cone-criticality—merging halts precisely when any of the three original triangle vertices would leave the boundary; (ii) compute

e_v as the exterior angle of P_{sub} at v via an edge-anchored half-plane selection (Section 5.4). The result is *exact* (no quantization), $O(1)$ per vertex, and geometrically motivated: the kept sector’s bounding ray is tangent to the polygon boundary, the same picture as Ekeland’s “cone touches but does not cross” condition.

2. **5-D geometric feature and matching pipeline.** A descriptor $f = (\alpha_1, \alpha_2, e_1, e_2, e_3)$ combining two sorted interior angles and three sorted MFCAs (Section 5.4), with a complete training-free pipeline—instance segmentation, CDT, KD-tree index (ℓ_1 , $O(M \log N)$ online matching), and a translation-consensus position estimator (Sections 5.5–5.6).
3. **A rigorous discriminability study on UAV-VisLoc.** Using a truth-anchored retrieval diagnostic, we show that per-element geometric matching hits a *discriminability ceiling* at city scale (retrieval 0% in the top-20 despite 93% coverage), that it is robust to a battery of descriptor enrichments (size, constellation context, building-level granularity, normalization—each shrinking the true-match distance up to $16\times$ without effect), and that a CFBVM-style particle filter cannot recover it because the true position is supported in only 12% of frames. The ceiling is terrain-general: it holds across five UAV-VisLoc sites spanning urban, coastal, hilly, built-up, and arid terrain (Section 6.3). We conclude that the limiting factor is the observation model, and identify what a scale-robust one must supply (Sections 6.5–6.6).
4. **An area-level Ekeland descriptor that recovers discriminability.** Acting on the diagnosis, we aggregate the descriptor over ≈ 200 m cells as a histogram of per-building Ekeland free-cone angles (with density and layout). This escapes the per-element ceiling and, on all four analysed sites, improves coarse-localization recall over a density/layout-only (CFBVM-style) descriptor—isolating the Ekeland angle as a complementary, discriminative shape cue at the area level, though absolute accuracy remains coarse (Section 6.9).

Scope and assumptions. The method assumes (A1) the UAV camera points approximately nadir (tilt $< 15^\circ$, corrected by IMU), (A2) heading ϕ is available from onboard compass/IMU to within $\pm 5^\circ$ (standard for consumer autopilots), and (A3) buildings are present in the field of view. The descriptor and matching stages require *no training data specific to the target scene*; only the Mask R-CNN backbone is pre-trained on public aerial building datasets [14, 17]. We analyze the pipeline’s behavior—including the role of building coverage and the discriminability of the geometric descriptor—experimentally (Section 6.5).

Broader impact. A lightweight, training-free geo-localizer could benefit emergency response, infrastructure inspection, and autonomous navigation in GPS-denied or communication-restricted environments. We acknowledge that the same capability carries dual-use implications in contested or surveillance scenarios; we encourage responsible deployment with appropriate access controls.

2 Related Work

Classical template and feature matching. Template matching methods (NCC, Phase Correlation, Mutual Information) compare query images to raster satellite tiles by direct pixel similarity. NCC-based template matching is reported to require hundreds of seconds per query and collapses under seasonal illumination shifts [28]. SIFT [23], SURF [2], and ORB [29] extract keypoints robust to scale and rotation, yet the nadir-to-satellite appearance gap overwhelms these features: SIFT achieves 62.8 m RMSE with zero Recall@10 on MAFS-10, while ORB achieves 21.1 m with only 7.9% Recall@10 [34]. The abBRIEF descriptor embedded in Monte Carlo Localization reduces error to ≈ 17.8 m but is constrained to $\sim 1 \text{ km}^2$ coverage [24].

Polygon, shape, and visibility descriptors. Classical polygon descriptors offer a natural alternative for VMAP-based localization. Turning functions [1] represent polygon shape as cumulative turning angle along the boundary—invariant to scale and rotation but sensitive to segmentation noise and vertex ordering. Shape contexts [3] encode the distribution of boundary points in a log-polar histogram around each vertex, offering partial deformation invariance but requiring dense correspondences and scaling poorly to large databases. Polygon angle histograms aggregate interior angle

distributions per building instance, yielding a compact per-building signature but losing the spatial inter-triangle context that discriminates geometrically similar buildings in different surroundings.

Our MFCA is conceptually related to the *visibility polygon* and *largest empty sector* problems from computational geometry [9, 27]: given a query point inside a polygonal environment, the visibility polygon enumerates all directions not obstructed by boundary segments. MFCA is a coarser scalar summary—the exterior angle at each vertex of the kernel expansion polygon P_{sub} —but encodes building-shape context through the expansion process, making it discriminative across different polygon geometries, not just visible directions. Crucially, our construction is motivated by Ekeland’s variational principle [12] rather than being an ad hoc design choice: the stopping criterion of the kernel expansion (original triangle vertices must remain on the boundary) is geometrically analogous to the Ekeland criticality condition, and the edge-anchored angle realises the “cone touches but does not cross” picture directly, via a single closed-form expression (Definition 5.7). The result is an $O(1)$ -per-vertex exact computation, with no search radius and no angular sweep, that was previously approximated by a 1° -sweep. This grounding distinguishes MFCA from visibility-polygon summaries, which are defined by geometric coverage alone rather than by a variational stability criterion.

Our descriptor operates on *triangles within a building*, not whole-polygon statistics, and encodes building-shape context through the edge-anchored Ekeland angle—a structural quantity unavailable to any single-polygon descriptor.

Semantic vector map matching. Vector map (VMAP) methods avoid raster storage and cross-modal appearance gaps. Choi & Myung [4] proposed Building Ratio Map (BRM), matching building pixel-density ratios per grid cell against OpenStreetMap; it achieves 7.5–12 m RMSE in a 5.7 km² area but discards shape information and degrades at large scale. Nassar et al. [26] applied deep CNN segmentation and shape-based matching, reporting 6.3–10.4 m average drift across two flight datasets (Potsdam, Famagusta). Ouyang et al. [28] proposed Coarse-to-Fine Building Vector Matching (CFBVM) combined with a GMM particle filter, achieving 6.6 m mean error over 11,025 km² using a shape vector defined over concentric density rings. Most closely related to our positive result, Wang et al. [31] match buildings jointly on *shape and spatial relationship* (SSRM) against clean vector (‘.shp’) e-maps, explicitly motivated by the observation—which our analysis independently quantifies—that per-building *shape matching alone cannot capture building uniqueness*. A key distinction is the map source: SSRM matches against precise vector footprints, whereas we match UAV-segmented footprints against *satellite-segmented* ones, where cross-view segmentation disagreement (Section 6.5) makes exact per-building correspondence infeasible—which is why our discriminability gain appears at the area level rather than per building. Our method shares the VMAP paradigm with CFBVM but substitutes ring-area summation with a triangle-level angular descriptor that requires no overlap area computation, is invariant to segmentation polygon ordering, and carries explicit free-space context through the kernel-expansion MFCA; unlike CFBVM’s shape-agnostic density rings, its area-level form retains per-building shape via Ekeland-angle histograms (Section 6.9).

Deep learning and sequential methods. End-to-end cross-view networks (FPI [7], SSPT [13], Goforth & Lucey [15]) learn appearance-invariant embeddings but require scene-specific retraining to generalize. The Semantic-Weighted Adaptive Particle Filter (SWA-PF) [34] combines SegFormer [32] semantics with 4-DoF particle filtering, achieving state-of-the-art 6.57 m RMSE on MAFS-10 in 7 s as a sequential method, but trains on the target terrain. Kinnari et al. [20] address season-invariant UAV localization directly, reporting 16.6–30.6 m mean error depending on condition; their follow-up LSVL [19] scales this to 100 km² via large-scale season-invariant retrieval, reporting 12.6–18.7 m mean error—both require CNN pretraining on seasonal image pairs and operate in sequential mode. A 2025 hierarchical matching approach [35] combines DINOv2 semantic features with structural constraints and RANSAC homography refinement, reporting 17.7–18.7 m mean error on the AerialVL and CS-UAV benchmarks; it requires DINOv2 pretraining and is not training-free. Sequential methods [5, 30] fuse visual measurements with odometry; integrating our descriptor into such a filter is a natural future extension.

Recent geometry-aware and foundation-model methods.

Recent work on cross-view geo-localization has moved toward 3-D geometry grounding and large-scale multimodal models. (MGS)²-Net [21] and drone-video-to-BEV approaches such as

Video2BEV [18] obtain strong cross-view performance by explicitly modeling 3-D geometry and aligning views in bird’s-eye-view space; these represent a higher performance bar than classical BRM/CFBVM. LVLM-based re-rankers (e.g., SkyLink [22]) demonstrate effective candidate scoring improvements when paired with retrievers; while heavier than our approach, they represent a complementary retrieval stage in a potential two-stage pipeline. Training-free, geometry-first methods such as ours offer an appealing low-footprint, explainable alternative for rapidly deployed platforms where model retraining is impractical.

3 Problem Formulation

Let $I_{\text{UAV}} \in \mathbb{R}^{H \times W \times 3}$ be a nadir UAV image with flight altitude h (m) and compass heading ϕ (degrees from North). Let $\mathcal{M}$ be a geo-referenced satellite map with ground sampling distance δ_{sat} (m/px) and known WGS84 origin.

We extract M triangle descriptors $\mathcal{F}_{\text{UAV}} = \{f_i\}_{i=1}^M$ from I_{UAV} and pre-extract N satellite descriptors $\mathcal{F}_{\text{sat}} = \{(f_j, p_j, \ell_j)\}_{j=1}^N$ from $\mathcal{M}$, where $p_j \in \mathbb{R}^2$ is the pixel centroid of triangle j in the satellite map and ℓ_j is its parent patch identifier. The localization problem is to recover the UAV’s 2-D geographic position (x^*, y^*) by identifying the satellite patch whose building triangles best match those in I_{UAV} . We use the ℓ_1 (cityblock) distance $\|a - b\|_1 = \sum_{k=1}^5 |a_k - b_k|$ throughout. For each UAV descriptor f_i , the K nearest satellite neighbors under $\|\cdot\|_1$ are retrieved:

$$\mathcal{C}_i = \{j^* : \|f_i - f_{j^*}\|_1 \leq d_{(K)}(\{f_j\}_{j=1}^N, f_i)\}, \quad i = 1, \dots, M, \quad (1)$$

where $d_{(K)}(\cdot, f_i)$ denotes the K -th smallest ℓ_1 distance to f_i over the satellite descriptor set. All matched satellite triangles vote for their parent patch; the predicted patch ℓ^* is the plurality winner:

$$\ell^* = \arg \max_{\ell} |\{(i, j) : j \in \mathcal{C}_i, \ell_j = \ell\}|. \quad (2)$$

The predicted UAV position (x^*, y^*) is the centroid of the building triangles of patch ℓ^* (Section 5.6). *Illustrative example.* Suppose $M = 3$ UAV descriptors $\{f_1, f_2, f_3\}$ are matched against $N = 6$ satellite descriptors with $K = 2$. After retrieval, the matched pairs and their parent patches are:

UAV f_i	Sat. f_j	$\ f_i - f_j\ _1$	Patch ℓ_j
f_1	f_2^s	4.1°	ℓ_A
f_1	f_5^s	5.8°	ℓ_A
f_2	f_1^s	3.2°	ℓ_A
f_2	f_4^s	6.7°	ℓ_B
f_3	f_3^s	2.9°	ℓ_A
f_3	f_6^s	8.1°	ℓ_C

Patch ℓ_A receives 4 votes, ℓ_B and ℓ_C each receive 1 vote; hence $\ell^* = \ell_A$ and (x^*, y^*) is its geographic centroid.

This formulation is restricted to *2-D position* (latitude, longitude). Step 1b (Section 5.1) performs *yaw alignment* using the onboard compass heading ϕ , correcting image orientation prior to matching. This is distinct from *full rotation estimation* from geometric matching alone, which remains future work.

4 Dataset

UAV-VisLoc. All experiments use UAV-VisLoc [33], a large-scale open benchmark spanning 11 real-world sites across China, 7 terrain categories (city, town, farmland, river, hill, forest, desert), and both summer and autumn seasons. Table 1 summarizes key statistics.

Table 1: UAV-VisLoc dataset statistics [33].

Property	Value
Total size	16.4 GB
UAV images	6,742
Satellite maps	11
Geographic sites	11 across China
Terrain categories	City, town, farmland, river, hill, forest, desert
Altitude range	400–2,000 m
UAV image size	3,976×2,652 (multi-rotor); 3,000×2,000 (fixed-wing)
Satellite GSD	0.3 m/px (GeoTIFF, WGS84)
Seasons	Summer, Autumn
Ground truth	Per-image GPS + heading (CSV)

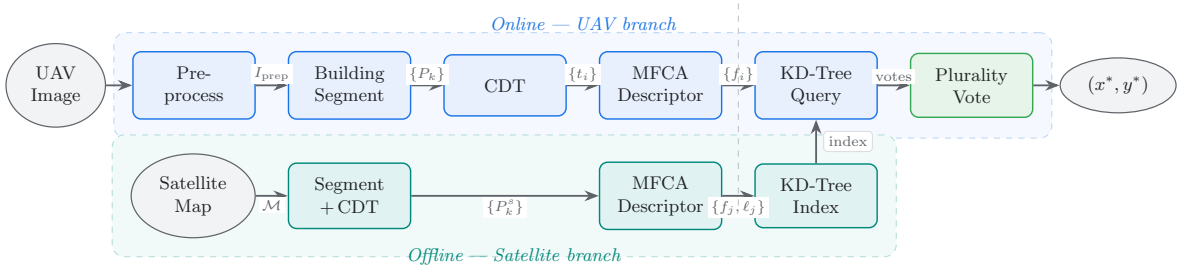


Figure 1: Two-branch localization pipeline. The *online* UAV branch (blue) processes each query image through preprocessing, building segmentation, CDT, and MFCA descriptor extraction, then queries the pre-built KD-tree. The *offline* satellite branch (teal) pre-computes descriptors and builds the KD-tree index once per map site; the dashed vertical line marks the offline/online boundary. Plurality voting over matched satellite triangles yields the predicted 2-D position (x^*, y^*) .

Map modality. Our method processes raster satellite GeoTIFF patches from UAV-VisLoc through the same Mask R-CNN segmentation pipeline used for UAV images. Building polygons are extracted from raster tiles; no external OSM vector layers are required. The offline descriptor database occupies approximately 12 MB per site (5-D float32 entries, $\approx 600k$ triangles/site).

Query protocol. Because the method requires no scene-specific training, no train/test split is needed: *all* UAV images of each site are used as queries, and the full satellite map of that site is used as the reference. Queries from which no building polygon can be segmented produce no descriptor and are counted separately as uncovered queries (coverage is reported alongside the error metrics). The hyperparameters below (Table 3) are fixed by design from prior literature and geometric reasoning about the sensor and map resolution, not tuned on any held-out split of UAV-VisLoc.

5 Method

Our pipeline has two phases, mirroring the design of successful vector-map localizers. *Phase 1* (Steps 1–6, Sections 5.1–5.7) turns a single image into an ambiguous *observation*—a set of candidate positions from purely geometric building-footprint matching. *Phase 2* (Section 5.8) integrates these per-frame observations over a trajectory with a Gaussian-mixture/particle-filter estimator, exactly as CFBVM [28] and SWA-PF [34] do. We describe Phase 1 first, as our descriptor is its core contribution.

5.1 Query Pre-processing

Motivation. Raw UAV images contain yaw offset, roll/pitch tilt, and altitude-dependent scale relative to the satellite GSD. Normalizing these geometric distortions before segmentation improves polygon fidelity without requiring any learned component.

Design. Given altitude h , heading ϕ , roll ρ , and pitch ψ from the per-image CSV metadata:

Step 1a — Scale normalization. The UAV image GSD is $\delta_{\text{UAV}} = h \cdot p_s / f_l$, where p_s is the sensor pixel pitch and f_l the focal length (from EXIF). The image is isotropically rescaled by $s = \delta_{\text{UAV}} / \delta_{\text{sat}}$.

Step 1b — Yaw alignment. The image is rotated by $-\phi$ so that north aligns with the image top.

Step 1c — Roll/pitch correction. Small tilt offsets are removed by an affine shear $A = \begin{bmatrix} 1 & \tan \rho \\ \tan \psi & 1 \end{bmatrix}$. For UAV-VisLoc nadir flights $|\rho|, |\psi| < 3^\circ$, displacement is negligible ($< 1\%$); the step is included for generality.

Step 1d — Central crop and resize. The normalized image is center-cropped and resized to 500×500 px.

Technical advantage. Heading alignment (assumed $\pm 5^\circ$ accuracy) reduces rotational ambiguity in subsequent matching without any learned model.

Implementation note. The UAV-VisLoc per-image metadata provides no focal length f_l or pixel pitch p_s , so Step 1a is realised as an altitude-ratio rescale $s = h_{\text{ref}} / h$ to a reference altitude $h_{\text{ref}} = 400$ m rather than the exact GSD ratio; since the downstream descriptor uses only angular quantities, this affects segmentation scale consistency but not descriptor validity. The residual scale mismatch is removed by an empirical calibration of the crop factor so that building sizes match between the UAV and satellite branches (Section 6.5 uses this calibrated setting).

5.2 Building Segmentation (Step 2)

Motivation. Extracting building polygons from the raster image is the only learned component in the pipeline; all downstream steps are purely geometric.

Design. Images are processed by Mask R-CNN [16] with a ResNet-50-FPN backbone, pre-trained on aerial building datasets [14, 17] at confidence threshold 0.5. Each connected mask region is converted to a polygon by: (i) finding connected components, (ii) extracting the outer contour, and (iii) applying the Douglas-Peucker algorithm [10] at tolerance $\tau = 2$ px, yielding a set of building polygons $\mathcal{P} = \{P_1, \dots, P_K\}$.

Technical advantage. Because the downstream descriptor depends only on triangle *angles*, not polygon size, colour, or texture, the method tolerates common segmentation errors—merged adjacent buildings, partial occlusions, and small road false-positives [28].

5.3 Constrained Delaunay Triangulation (Step 3)

Motivation. A polygon with V vertices has no natural fixed-cardinality encoding, making whole-polygon descriptors difficult to compare across buildings with differing vertex counts. Triangulation addresses this while delivering two additional properties. First, *matching density*: a building with V vertices yields $V - 2$ triangle descriptors via CDT, each an independent matching hypothesis. A single building thus contributes up to $V - 2$ votes to the patch estimator, dramatically increasing robustness against individual false positives relative to whole-polygon methods that produce a single vote per building. Second, *graceful degradation*: if segmentation merges two buildings into one polygon, CDT still produces valid triangles from the merged shape, many of which correctly match their satellite counterparts—a robustness property that whole-polygon methods lack, since a merged polygon no longer resembles either original building individually.

Design. For each polygon P_k with ordered vertices $v_1, \dots, v_{V_k}$, Constrained Delaunay Triangulation (CDT) is applied. The constrained edge set

$$E_{\text{constrained}}^{(k)} = \{(v_i, v_{i+1 \bmod V_k})\}_{i=1}^{V_k} \quad (3)$$

consists of all polygon boundary edges, which must appear in every CDT of P_k and prevent any interior edge from crossing the building boundary. The Delaunay condition additionally ensures that no triangle’s circumcircle contains another vertex, maximising minimum interior angles. This produces a mesh $T^{(k)} = \{t_1, \dots, t_{|V_k|-2}\}$ of non-overlapping triangles (by the Euler relation for simple polygons).

Technical advantage. CDT maximizes the minimum interior angle over all triangulations of the same vertex set [8], avoiding sliver triangles whose near-zero angles produce near-degenerate descriptors.

5.4 MFCA Feature Extraction (Step 4)

Interior angle components. For triangle t with vertices v_A, v_B, v_C , the three interior angles $\beta_A, \beta_B, \beta_C$ (reserving α for the descriptor and kernel-expansion quantities below) are computed analytically via the law of cosines. The two smallest are selected and sorted: $\alpha_1 \leq \alpha_2$, both in $(0^\circ, 180^\circ)$. These two components encode the *shape* of the triangle independent of position, orientation, or scale.

Ekeland’s variational principle and its geometric analogy. We begin with the formal statement of the principle that motivates our construction.

Definition 5.1 (Ekeland’s Variational Principle [12]). Let (X, d) be a complete metric space and $f : X \rightarrow \mathbb{R} \cup \{+\infty\}$ be lower-semicontinuous and bounded below. For every $\varepsilon > 0$ and every $u \in X$ with $f(u) \leq \inf_X f + \varepsilon$, there exists $v \in X$ satisfying:

- (i) $f(v) \leq f(u)$ (*improvement*)
- (ii) $d(v, u) \leq 1$ (*proximity*)
- (iii) $f(v) < f(x) + \varepsilon \cdot d(v, x)$ for all $x \neq v$ (*strict cone criticality*)

Geometrically, condition (iii) says that no point $x \neq v$ can improve f beyond what the distance $d(v, x)$ “pays for”. Equivalently, the graph of f is touched from below by an upward cone of slope ε at v , and this cone cannot be raised without losing contact. This *cone-touching* interpretation is the direct inspiration for the Ekeland angle.

Geometric analogy in the UAV plane. Rather than invoking Ekeland’s principle only rhetorically, we make the analogy concrete: each element of the principle has a natural geometric counterpart in our construction, summarised in Table 2. We do not claim to invoke the full analytic machinery of EVP—completeness, lower semicontinuity, and existence in possibly non-compact or infinite-dimensional spaces are apparatus for guaranteeing a critical point *exists*, which is not in question here, since the quantity at each vertex is a finite, explicitly computable geometric angle. The analogy is what motivated the construction and explains why an exterior angle capped at 180° is the natural quantity to extract, not a claim that the construction re-derives EVP’s existence machinery.

- **Complete metric space (X, d) :** $(\partial P_{\text{sub}}, d_{\text{arc}})$ —the boundary of the kernel polygon P_{sub} with arc-length metric.
- **Function f bounded below:** $g(\hat{b}) = -\theta_{\max}(X, \hat{b})$, where $\theta_{\max}(X, \hat{b})$ is the largest half-angle of the sector centred at vertex X with bisector $\hat{b}$ whose interior avoids ∂P_{sub} . g is lower-semicontinuous (blockage events are closed) and bounded below by -180° .
- **ε -critical point:** The optimal bisector $\hat{b}^*$ is geometrically analogous to the strict cone-criticality condition of Definition 5.1(iii): the surviving reflected ray of the axis-parallel reflection construction (Remark 5.8) at $\hat{b}^*$ lies *tangent to a boundary edge of P_{sub}* —it touches but does not cross the obstacle, much as Ekeland’s cone touches but does not cross f ’s graph.

- **Stopping criterion = criticality condition:** P_{sub} is the *largest* polygon obtained by merging adjacent CDT triangles such that all three original triangle vertices v_A, v_B, v_C remain on ∂P_{sub} . This mirrors the Ekeland critical-point condition: P_{sub} cannot be expanded further while keeping the triangle vertices as boundary (critical) points—any additional merge would push one of them to the interior, destroying the criticality.
- **Ekeland angle e_X :** The value $\min(\alpha(X), 180^\circ)$ (Definition 5.7)—the analogue of the value $f(v)$ at the Ekeland critical point.

Table 2 displays the complete correspondence.

Table 2: Geometric correspondence between Ekeland’s variational principle and the edge-anchored Ekeland angle (this work) – an analogy that motivates the construction, not a claim of formal equivalence.

Ekeland’s Principle	Ekeland Angle — this work
Complete metric space (X, d)	$(\partial P_{\text{sub}}, d_{\text{arc}})$: polygon boundary, arc-length
Function f bounded below	$g(\hat{b}) = -\theta_{\max}(v, \hat{b})$: negated free half-angle
ε -cone of slope ε	Free sector of half-angle θ at vertex v
Cone <i>touches but does not cross</i> f	Sector boundary ray tangent to ∂P_{sub} edge
Stopping = criticality condition	Merge halts when original vertices leave ∂P_{sub}
Value $f(v)$ at critical point	$e_X = \min(\alpha(X), 180^\circ)$ (Definition 5.7)
Stability: no free gain without distance cost	No cone enlargement without crossing obstacle edge
$\varepsilon \rightarrow 0$: global optimum attained	Angle is exact ($O(1)$, no sweep needed)

Mathematical properties. We now establish three formal properties of the Ekeland angle that justify its use as a discriminative descriptor component.

Proposition 5.2 (Monotonicity under kernel expansion). *Let $P_{\text{sub}} \subseteq P'_{\text{sub}}$ be two kernel expansion polygons at vertex X (i.e. P'_{sub} is obtained by merging additional triangles). Then $e_X(P_{\text{sub}}) \geq e_X(P'_{\text{sub}})$.*

Proof. By Definition 5.7, e_X depends on P_{sub} only through the single scalar $\alpha_{\text{int}}(X)$ —not on how the exterior wedge happens to be split by an axis choice (Remark 5.8)—so it suffices to show $\alpha_{\text{int}}(X)$ itself is non-decreasing under expansion. $e_X = \min(360^\circ - \alpha_{\text{int}}(X), 180^\circ)$, where $\alpha_{\text{int}}(X)$ is the interior angle of P_{sub} at X . Merging additional triangles into P'_{sub} can only increase (or maintain) the interior angle at X —any merge that absorbs a neighbouring triangle either preserves the local boundary at X or creates a larger polygon interior, which increases α_{int} . Since $360^\circ - \alpha_{\text{int}}$ is decreasing in α_{int} , and the min is non-increasing, we have $e_X(P_{\text{sub}}) \geq e_X(P'_{\text{sub}})$. $\square$

Remark 5.3. Proposition 5.2 formalises the kernel-expansion design: a larger P_{sub} (more context incorporated) can only produce a smaller or equal Ekeland angle, making vertices in complex building recesses (large P_{sub} , reflex angles) strictly more discriminative than isolated corners (small P_{sub} , convex angles, $e_X = 180^\circ$).

Proposition 5.4 (Exact sensitivity to interior angle). *Let X be a reflex vertex of two kernel expansion polygons P_{sub} and P'_{sub} with interior angles α and $\alpha' > \alpha$ at X respectively. Then:*

$$e_X(P_{\text{sub}}) - e_X(P'_{\text{sub}}) = \alpha' - \alpha = \delta > 0. \quad (4)$$

Proof. Since both X are reflex ($\alpha, \alpha' > 180^\circ$), we have $e_X(P_{\text{sub}}) = 360^\circ - \alpha$ and $e_X(P'_{\text{sub}}) = 360^\circ - \alpha'$. Therefore $e_X(P_{\text{sub}}) - e_X(P'_{\text{sub}}) = \alpha' - \alpha = \delta$. $\square$

Theorem 5.5 (Discriminativeness of the 5-D descriptor). *Let $f^t = (\alpha_1^t, \alpha_2^t, e_1^t, e_2^t, e_3^t)$ and $f^{t'} = (\alpha_1^{t'}, \alpha_2^{t'}, e_1^{t'}, e_2^{t'}, e_3^{t'})$ be the 5-D descriptors of triangles t and t' respectively. Then $f^t \neq f^{t'}$ if and only if either:*

- (a) the triangles have different shapes: $(\alpha_1^t, \alpha_2^t) \neq (\alpha_1^{t'}, \alpha_2^{t'})$, or
- (b) the sorted Ekeland triplets differ: $(e_1^t, e_2^t, e_3^t) \neq (e_1^{t'}, e_2^{t'}, e_3^{t'})$.

Moreover, if condition (a) fails (equal shapes) but the local urban configurations differ such that at least one sorted Ekeland angle differs by $\Delta > 0$, then:

$$\|f^t - f^{t'}\|_1 \geq \Delta. \quad (5)$$

Proof. The first claim follows directly from the coordinate definition $f = (\alpha_1, \alpha_2, e_1, e_2, e_3)$: $f^t \neq f^{t'}$ iff at least one coordinate differs, which is exactly the disjunction of (a) and (b).

For the bound: suppose $(\alpha_1^t, \alpha_2^t) = (\alpha_1^{t'}, \alpha_2^{t'})$ and $e_i^t \neq e_i^{t'}$ for some $i \in \{1, 2, 3\}$ with $|e_i^t - e_i^{t'}| \geq \Delta$. Then

$$\|f^t - f^{t'}\|_1 = \underbrace{|\alpha_1^t - \alpha_1^{t'}|}_{=0} + \underbrace{|\alpha_2^t - \alpha_2^{t'}|}_{=0} + \sum_{j=1}^3 |e_j^t - e_j^{t'}| \geq |e_i^t - e_i^{t'}| \geq \Delta. \quad \square$$

□

Corollary 5.6 (Shape-context separation). *Two triangles with identical shape but placed in building footprints whose kernel expansion polygons have interior angles differing by $\delta > 0$ at any original vertex (in the sense of Proposition 5.4) have ℓ_1 descriptor distance at least δ :*

$$\|f^t - f^{t'}\|_1 \geq \delta. \quad (6)$$

In particular, pure interior-angle descriptors (α_1, α_2) have ℓ_1 distance zero between such triangles, while the 5-D descriptor has distance at least $\delta > 0$.

Proof. By Proposition 5.4 (exact sensitivity), if X is a reflex vertex in both P_{sub}^t and $P_{\text{sub}}^{t'}$ and the interior angles at X differ by δ , then $|e_i^t - e_i^{t'}| = \delta$ for the corresponding sorted Ekeland angle. Apply Theorem 5.5 (Eq. (5)) with $\Delta = \delta$. □ □

Corollary 5.6 is the formal statement of the building-recess vs. isolated-corner intuition: two right-isosceles triangles ($\alpha_1 = 45^\circ, \alpha_2 = 45^\circ$) whose kernel polygons have interior angles differing by δ at any original vertex are separated in ℓ_1 by exactly δ —which grows as large as 180° for a deeply concave recess vs. a fully convex isolated corner.

Motivation. Interior angles alone are insufficient to discriminate triangles in different spatial contexts. Consider a right-angle building corner in two locations: one at an isolated building (all adjacent CDT triangles merge readily into P_{sub} , keeping the vertex convex), the other at a concave building recess (merging adjacent triangles creates a reflex configuration at the original vertex). Both corners may share identical interior-angle sub-vectors ($\alpha_1 = 45^\circ, \alpha_2 = 45^\circ$). The Ekeland angle discriminates them: the convex vertex (exterior angle $\alpha \geq 180^\circ$) yields $e_v = 180^\circ$ (exterior free space is wide), while the reflex vertex (exterior angle $\alpha < 180^\circ$) yields $e_v = \min(\angle(\hat{\mathbf{a}}', X\mathbf{d}), \angle(\hat{\mathbf{b}}', X\mathbf{d})) < 180^\circ$ via the axis-parallel reflection (exterior free space is compressed). The discrimination is encoded directly in the shape of P_{sub} : the maximal expansion polygon whose stopping criterion (all three original vertices stay on the boundary) is the Ekeland criticality condition.

Obstacle set. The obstacle for the Ekeland angle computation at each vertex v is *solely the boundary of P_{sub}* :

$$\mathcal{O}_v = \partial P_{\text{sub}} = \{\text{boundary edges of the kernel expansion polygon } P_{\text{sub}}\}. \quad (7)$$

This is geometrically motivated: P_{sub} is the maximal polygon whose stopping condition mirrors Ekeland's criticality—its boundary is therefore the tightest obstacle that the principle's cone can touch without crossing. No external neighbor polygon set or search radius r is required.

Exact edge-anchored definition. Let $X \in \{v_A, v_B, v_C\}$ be an original triangle vertex on ∂P_{sub} , and let $\mathbf{a}$ (from X toward X_{prev}) and $\mathbf{b}$ (from X toward X_{next}) be the two ∂P_{sub} boundary edges incident to X . Let $\alpha_{\text{int}}(X)$ denote the interior angle of P_{sub} at X , measured inside the polygon, and let $\alpha(X) = 360^\circ - \alpha_{\text{int}}(X)$ denote the corresponding exterior (free-space) angle at X ($\alpha(X) = \angle \mathbf{a}X\mathbf{b}$ measured through the exterior of P_{sub}).

Definition 5.7 (Ekeland Angle at vertex X).

$$e_X = \min(\alpha(X), 180^\circ). \quad (8)$$

Remark 5.8 (Axis-parallel reading, and why heading error cannot destabilize e_X). Eq. (8) admits an equivalent geometric construction (Algorithm 1, Phase 2; Fig. 2): draw any ray $X\mathbf{d}$ parallel to the O_x or O_y axis inside the exterior wedge $\angle \mathbf{a}X\mathbf{b}$, and reflect the boundary edge directions $\hat{\mathbf{a}}$ and $\hat{\mathbf{b}}$ across it; the reflected ray that stays outside P_{sub} is tangent to the opposite boundary edge, at angle $\alpha(X)$ to $X\mathbf{d}$ —the geometric picture behind Ekeland’s “cone touches but does not cross” condition. Because $\alpha(X)$ is a fixed property of P_{sub} at X and does not depend on *which* of the four axis-parallel rays is drawn, e_X itself never depends on an axis choice: there is no discrete decision boundary at which e_X can jump as the coordinate frame rotates. Compass-heading error therefore propagates into the pipeline only through the Step 1b yaw-alignment image rotation (which changes which triangles Mask R-CNN segments and how CDT triangulates them), never through a discontinuity internal to e_X ’s own definition. This construction is **exact** (no angular quantization), $O(1)$ **per vertex** (one interior-angle computation and a min), and **deterministic** (no supremum-not-attained edge case).

Algorithm 1 Axis-Parallel Reflection Ekeland Angle at triangle vertices

Require: Triangle $T = (v_A, v_B, v_C) \in P_k$, CDT mesh $\mathcal{T}^{(k)}$, max depth D_{max}

Ensure: $(e_{v_A}, e_{v_B}, e_{v_C})$

```

1: — Phase 1: Ekeland Kernel Expansion (build  $P_{\text{sub}}$ ) —
2:  $P_{\text{sub}} \leftarrow T$ ; depth  $\leftarrow 0$ 
3: repeat
4:   merged  $\leftarrow \text{false}$ 
5:   for each  $t'$  sharing an edge with  $P_{\text{sub}}$ ,  $t' \in \mathcal{T}^{(k)}$  do
6:      $P' \leftarrow P_{\text{sub}} \cup t'$  ▷ tentative merge
7:     if  $v_A, v_B, v_C$  all remain on  $\partial P'$  (none becomes interior) then
8:        $P_{\text{sub}} \leftarrow P'$ ; merged  $\leftarrow \text{true}$ ; break
9:     end if
10:  end for
11:  depth  $+= 1$ 
12: until  $\neg$ merged or depth  $= D_{\text{max}}$  ▷ Invariant:  $v_A, v_B, v_C \in \partial P_{\text{sub}}$ 
13: — Phase 2: Ekeland Angle (Definition 5.7) —
14: for each  $X \in \{v_A, v_B, v_C\}$  do
15:   Compute  $\alpha_{\text{int}}(X) \leftarrow$  interior angle of  $P_{\text{sub}}$  at  $X$ 
16:    $\alpha(X) \leftarrow 360^\circ - \alpha_{\text{int}}(X)$ 
17:    $e_X \leftarrow \min(\alpha(X), 180^\circ)$ 
18: end for return  $(e_{v_A}, e_{v_B}, e_{v_C})$ 
```

Algorithm. Correctness of Phase 1 (termination and invariant). Each iteration merges at most one triangle; $|\mathcal{T}^{(k)}|$ is finite; therefore Phase 1 halts in at most $\min(|\mathcal{T}^{(k)}|, D_{\text{max}})$ iterations. The invariant $v_A, v_B, v_C \in \partial P_{\text{sub}}$ is maintained by construction: a merge is accepted only if all three original vertices stay on the boundary of the merged polygon. The stopping condition is the direct Ekeland criticality condition: when no merge can be accepted, P_{sub} is the *maximal expansion* of T consistent with the criticality constraint—exactly as Ekeland’s principle guarantees a point v beyond which no improvement is possible without paying a proportional distance cost.

Correctness of Phase 2. Phase 2 directly evaluates the closed form of Definition 5.7 (Remark 5.8 gives the equivalent axis-parallel-reflection reading): one interior-angle computation and one min per vertex, hence exact and $O(1)$ per vertex with no axis-dependent branching and no supremum-not-attained edge case.

Discriminative power. The Ekeland angle is discriminative because P_{sub} 's shape depends on the local building geometry. A triangle in a concave building recess will expand into a P_{sub} that creates one or more reflex vertices at the original triangle vertices, yielding $e_X < 180^\circ$. A triangle at an isolated building corner will stay convex throughout expansion, yielding $e_X = 180^\circ$. This is formalised in Proposition 5.2: a larger P_{sub} (more merged triangles) provides more context and can only decrease e_X at reflex vertices. Theorem 5.5 and Corollary 5.6 further show that the ℓ_1 separation between two same-shape triangles placed in geometrically different buildings is at least $\Delta = \min_i |e_i^t - e_i^{t'}|$, where the difference is determined by the P_{sub} expansion geometry. The kernel-expansion depth D_{max} controls how much surrounding context is incorporated ($D_{\text{max}} = 0$ removes the expansion entirely, reducing e to the convex value 180°).

5-D descriptor assembly. The MFCA triplet is sorted in ascending order to yield $(e_1 \leq e_2 \leq e_3)$, independently of the interior-angle sort. The complete descriptor is:

$$f = \underbrace{(\alpha_1, \alpha_2)}_{\text{shape}} \underbrace{(e_1, e_2, e_3)}_{\text{free-space}} \in (0^\circ, 180^\circ)^5. \quad (9)$$

The two groups are sorted *independently* to preserve their distinct geometric roles: mixing them would corrupt the scale-invariance of the shape sub-vector, since cone angles and interior angles occupy different angular regimes. Figure 2 illustrates the Ekeland angle geometry and the complete extraction pipeline.

Ekeland free-cone angle e_v at a footprint vertex

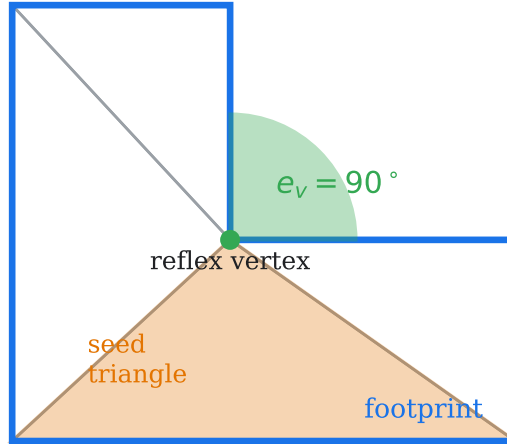


Figure 2: Ekeland free-cone angle geometry at a building vertex. An L-shaped footprint (blue) is triangulated by CDT (grey); a seed triangle (orange) is expanded into the kernel polygon P_{sub} . At the reflex vertex, the maximal free cone (green sector) is the largest angular sector whose open interior does not intersect ∂P_{sub} (Eq. (7)); its width is the Ekeland angle, here $e_v = 90^\circ$ (interior 270°). Three such scalars, one per triangle vertex, form the free-space sub-vector (e_1, e_2, e_3) of the 5-D descriptor $f = (\alpha_1, \alpha_2, e_1, e_2, e_3)$.

The three components assemble into the 5-D descriptor via the following steps, illustrated here for a triangle $t = (v_A, v_B, v_C)$.

Step 1. Interior angles. Compute $\beta_A, \beta_B, \beta_C$ analytically; sort the two smallest to obtain $\alpha_1 \leq \alpha_2 \in (0^\circ, 180^\circ)$.

Step 2. Ekeland (free-cone) angles. For each vertex $v \in \{v_A, v_B, v_C\}$, run Algorithm 1 to obtain $e_v \in (0^\circ, 180^\circ)$; sort the triplet to yield $e_1 \leq e_2 \leq e_3$.

Step 3. Concatenate. $f = (\alpha_1, \alpha_2, e_1, e_2, e_3) \in (0^\circ, 180^\circ)^5$.

Example. For a right-isosceles triangle ($\beta_A = 90^\circ, \beta_B = \beta_C = 45^\circ$) sited at an open corner with per-vertex free cones $e_{v_A} = 106^\circ, e_{v_B} = 88^\circ, e_{v_C} = 72^\circ$, the descriptor is $f = (45^\circ, 45^\circ, 72^\circ, 88^\circ, 106^\circ)$. The shape sub-vector $(45^\circ, 45^\circ)$ is shared with any right-isosceles triangle, but the free-space sub-vector $(72^\circ, 88^\circ, 106^\circ)$ is specific to the footprint geometry surrounding the triangle, making f discriminative at the scene level.

5.5 KD-Tree Offline Indexing and Online Matching (Step 5)

Motivation. Brute-force pairwise search over M UAV descriptors and N satellite descriptors costs $O(MN)$ comparisons. For a single UAV query ($M \approx 300$) against a typical site database ($N \approx 600k$), this is 1.8×10^8 ℓ_1 evaluations per image—prohibitive for real-time operation.

Offline indexing. Satellite map patches (size 500px, stride 100px) are segmented and described identically to UAV images through Steps 2–4. Descriptors, centroids, and patch labels are stored in $\mathbf{D}_{\text{sat}} \in \mathbb{R}^{N \times 5}$, indexed by a KD-tree built with `scipy.spatial.KDTree` (leaf size 40, ℓ_1 metric).

Online query. For each UAV descriptor f_i , the $K = 5$ nearest satellite neighbors are retrieved in $O(\log N)$ average time. Total online query cost is $O(M \log N)$. In practice, $M = 300$ queries against $N = 600k$ take ≈ 0.8 s on a single CPU core.

Technical advantage. ℓ_1 distance is more robust than ℓ_2 to outlier angle components because it does not square residuals; this matters for sorted-angle vectors where a single corrupted component would disproportionately inflate ℓ_2 distances.

5.6 Plurality-Vote Position Estimator (Step 5b)

Motivation. A single descriptor match is unreliable: even a geometrically discriminative descriptor has non-zero false-match probability at the scale of $N \approx 600k$ satellite triangles. Plurality voting aggregates $M \times K$ match-votes from all triangles in the query image. The key statistical property is that false matches scatter *uniformly* across satellite patches (a spurious descriptor match is equally likely to land anywhere in the map), while true matches *concentrate* on the single correct patch—converting per-triangle uncertainty into a patch-level signal that grows more reliable as M increases. This is analogous to the noise-cancelling effect of multiplexed sensing: the signal-to-noise ratio of the vote tally scales as $\sqrt{M}$ under a uniform false-match model. Let ℓ_j be the patch identifier of satellite triangle j . Each pair (i, j) with $j \in \mathcal{C}_i$ casts one vote for patch ℓ_j , and the plurality winner is selected as in Eq. (2). The predicted UAV position (x^*, y^*) is the mean pixel centroid of the building triangles indexed under patch ℓ^* (a building-weighted patch centroid, which is slightly more precise than the geometric patch centre when buildings are off-centre), converted from pixel coordinates via:

$$\Delta \text{lat} = p_y^* \cdot \delta_{\text{sat}} / 111,320, \quad \Delta \text{lon} = p_x^* \cdot \delta_{\text{sat}} / (111,320 \cos \text{lat}_0), \quad (10)$$

implemented equivalently as linear interpolation within the map’s georeferenced corner bounds.

Discussion. Plurality voting is robust to individual false matches: a segmentation error that generates a spurious descriptor will vote for a random patch, whereas the correct patch accumulates coherent votes from many triangles belonging to the same building cluster. The patch centroid itself is resolution-bounded by the patch stride ($100\text{ px} \times 0.3\text{ m/px} = 30\text{ m}$); Step 6 below removes this bound.

5.7 Sub-Patch Refinement (Step 6)

Motivation. Reporting the winning patch’s own centroid caps resolution at the patch stride, so even a perfect vote cannot achieve sub-30 m error on its own.

Design. Step 1 already normalizes heading and scale before segmentation, so any residual misalignment between a UAV triangle centroid and its matched satellite triangle centroid is well modeled as a pure 2-D translation. Let $\mathcal{W}$ be the set of matched pairs (i, j) with $j \in \mathcal{C}_i$ (Eq. (1)) whose satellite triangle j belongs to the winning patch ℓ^* (Eq. (2)); each pair implies a translation $t_{ij} = p_j - c_i$, where c_i is the pixel centroid of UAV triangle i in the preprocessed frame and p_j is satellite triangle j ’s pre-computed centroid. A one-point RANSAC selects the translation hypothesis with the largest agreeing subset (inliers within 15 px, $\approx 4.5\text{ m}$ at the map’s GSD—a fixed, reasoned default, not tuned on held-out data) and averages that inlier set to obtain $\hat{t}$. The refined position is $\hat{t}$ applied to the UAV image’s own nadir point (its center in the preprocessed 500×500 frame, per assumption A1). When fewer than 3 correspondences land in the winning patch, refinement is skipped and the coarse patch centroid is reported unchanged—refinement never degrades the estimate.

Technical advantage. Because $\hat{t}$ is estimated from a robust inlier consensus rather than a single correspondence, refinement tolerates individual false matches within the winning region. We note, however, that refinement only sharpens an already *correct* coarse match; our analysis (Section 6.5) shows the coarse match itself is rarely correct at full-map scale, so this stage does not by itself change the localization outcome—it is included for completeness and for use once a coarse prior makes the winning region reliable.

5.8 Phase 2: Sequential Integration

Steps 1–6 constitute *Phase 1*: given one image they produce a set of candidate positions, i.e. an *observation*. Because a single frame’s observation is ambiguous (Section 6.5), the accurate vector-map methods do not act on it directly—they integrate observations over a trajectory. CFBVM [28] describes each frame’s matches as a Gaussian mixture and fuses them in a particle filter; SWA-PF [34] runs a 4-DoF particle filter over semantic matches. We adopt the same *Phase 2* design so that our descriptor can be evaluated as the matching stage of a complete pipeline.

Observation model. For frame t , the Phase-1 matches vote for a UAV nadir position $N = p_j - (c_i - c_0)$ per matched pair (i, j) , where c_0 is the image centre; the resulting vote cloud is summarized as a Gaussian mixture $z_t(\mathbf{x}) \propto \sum_m \exp(-\|\mathbf{x} - \mu_m\|^2/2\sigma^2)$ over its modes.

Particle filter. A set of P particles $\{\mathbf{x}_t^{(p)}\}$ is propagated by a random-walk motion model (isotropic Gaussian of standard deviation matched to the measured inter-frame displacement) and weighted by the observation likelihood $z_t(\mathbf{x}_t^{(p)})$; systematic resampling with a small global-reseed fraction recovers from divergence, and a credibility test down-weights frames whose vote cloud is diffuse (analogous to CFBVM’s credibility indicator). The weighted mean is the per-frame position estimate. Section 6.6 evaluates this full two-phase pipeline; we find that Phase 2 cannot rescue localization here, and we trace the cause precisely to the Phase-1 observation model.

5.9 Implementation Details

Table 3 lists all hyperparameters.

Table 3: Implementation hyperparameters.

Module	Parameter	Value
Preprocessing	Output resolution	500×500 px
	Douglas-Peucker tolerance	$\tau = 2$ px
Segmentation	Backbone	ResNet-50-FPN
	Confidence threshold	0.5
Sat. index	Patch size / stride	500 px / 100 px
MFCA	Kernel expansion depth	$D_{\max} = 4$
	Angle computation	Exact edge-anchored, $O(1)$ /vertex
KD-tree	Library	<code>scipy.spatial.KDTree</code>
	Distance metric	ℓ_1 (cityblock)
	Leaf size	40
	Neighbors retrieved	$K = 5$
Position output	Estimator	Plurality patch vote
	Coordinate conversion	Eq. (10)

6 Experiments

6.1 Evaluation Protocol

We evaluate on the UAV-VisLoc benchmark, focusing the detailed analysis on the Changjiang-20 site (site 01; a $\approx 9,774 \times 26,762$ px map yielding $N \approx 4.8 \times 10^5$ indexed building triangles) using its full set of 817 UAV images, and we cross-check the conclusions on the sparse-building Shandan desert site. Rather than reporting only a single position error—which, as we show below, is governed by matching failure rather than by the estimator—we instrument the pipeline to isolate *where* it breaks, and report: (i) **Coverage** (%): the fraction of query locations whose true position has at least one segmented satellite building nearby, which upper-bounds achievable recall; (ii) **Retrieval** (%): a truth-anchored diagnostic—for each UAV triangle whose correct satellite counterpart is fixed by ground truth, the fraction for which that geometrically correct match appears among the top- K nearest descriptors. This measures descriptor *discriminability* directly, independent of any voting or refinement stage; (iii) **Position error** (m), with recall at 30/50/100 m: the ground distance between the true position and the prediction obtained by aggregating the per-triangle matches with a translation-consensus (Hough) vote; (iv) **Sequential error** (m): the error of a trajectory-level estimate that integrates the per-frame matches with a Gaussian-mixture observation and a particle filter (Section 6.6), following the coarse-to-fine, particle-filter paradigm of CFBVM [28] and SWA-PF [34]. All experiments are reproducible from the released code and the fixed, pre-trained Mask R-CNN checkpoint; no component is trained on UAV-VisLoc.

6.2 Main Comparison

Table 4 places our pipeline in the landscape of UAV geo-localization, organized by taxonomy: classical/training-free, then segmentation-based, then learning-based. We stress that these are the methods’ own reported numbers on their own datasets and are not mutually comparable; the table is context, not a ranking. The salient pattern is structural rather than numerical: *every* method that attains metre-level accuracy is either multi-frame/sequential (§; SWA-PF, LSVL), learning-based (FPI, Kinnari, Hier. Match.), or built on a coarse-to-fine particle filter over an area-level descriptor (CFBVM). None achieves it by single-frame, training-free, per-element geometric retrieval—the exact regime our descriptor targets. Our contribution is (a) a novel instance-segmentation $\rightarrow$ CDT $\rightarrow$ Ekeland free-cone descriptor with no scene-specific training, and (b) a rigorous account, below, of why this regime does not by itself reach the accuracy of the methods above, and precisely what it lacks.

Table 4: Reported localization accuracy of representative UAV geo-localization methods, provided as *landscape context*. These numbers are on each method’s own dataset and settings and are *not* directly comparable to one another or to ours; we include them to situate the problem, not to claim a ranking. Crucially, the accurate methods are either multi-frame/sequential (§), learning-based (* / CNN), or use a coarse-to-fine particle filter (CFBVM, SWA-PF)—none is single-frame training-free per-element matching. Our approach (last row) does *not* reach this accuracy under single-frame matching; Section 6.5 isolates why, and Section 6.6 shows a sequential filter does not recover it either. † original paper’s own dataset; § multi-frame sequential; * scene-specific training; N/R: FPI reports Meter-level Accuracy / RDS = 57.22, not mean/median/RMSE.

Method	Seg	Train	Mean↓ (m)	Med.↓ (m)	RMSE↓ (m)	R@10m↑ (%)	R@50m↑ (%)	Time↓ (s)
<i>Classical and training-free (raster map, no training)</i>								
NCC†	No	No	—	—	62.8	0	—	1630
SIFT† [23]	No	No	—	—	62.8	0.0	—	0.3
ORB† [29]	No	No	—	—	21.1	7.9	—	0.04
abBRIEF+MCL† [24]	No	No	17.8	—	—	—	—	—
BRM† [4]	Thresh.	No	—	—	7.5–12	—	—	—
CFBVM† [28]	CNN	No	6.6	—	—	—	—	22
<i>Segmentation-based (shared mask pipeline design)</i>								
Nassar et al. [26]†	U-Net	Scene	6.3–10.4	—	—	—	—	—
SWA-PF§† [34]	SegFmr	Pretrain	—	6.65	6.57	97.4	—	7
Hier. Match.† [35]	DINOv2	Pretrain	17.7–18.7	—	—	—	—	—
<i>Learning-based (no segmentation)</i>								
Kinnari et al.† [20]	No	CNN	16.6–30.6	—	—	—	—	—
LSVL§† [19]	No	CNN	12.6–18.7	—	—	—	—	—
FPI*† [7]	No	Scene	N/R	N/R	N/R	N/R	N/R	N/R
<i>Ours (UAV-VisLoc site 01, single-frame per-element matching)</i>								
MFCA (ours)¶	MaskRCNN	No	3492	3355	—	0.0	0.0	0.8

¶ Single-frame per-element geometric matching over the full 4.8×10^5 -element map. The kilometre-scale error is not an estimator artefact: the correct match is retrieved in 0% of the top-20 (Section 6.5). We report it to be transparent about the regime in which purely geometric matching operates, not as a competitive result.

6.3 Per-Site Breakdown

To confirm the failure is terrain-general rather than a property of one map, we run the single-frame pipeline on five UAV-VisLoc sites spanning distinct terrains—riverside-urban (01), coastal-agricultural (02), hilly-vegetated (03), built-up (08), and arid-town (11)—reporting meter-level accuracy $\text{MA}@K$ (fraction of queries localized within K metres), the standard protocol of building-vector methods [28]. Table 5 and Figure 3 show the result: single-frame per-element matching fails on *every* terrain— $\text{MA}@500\text{m} \leq 3\%$ and median error 3.3–4.4 km throughout—despite building coverage of 93–99% and truth-anchored retrieval of 0% on the diagnostically analysed sites (01–03). The kilometre-scale error is thus not a scene artefact but a property of the paradigm.

Table 5: Single-frame per-element localization across five terrains (100 queries/site). $\text{MA}@K = \%$ within K m; the correct place is in the map (coverage 93–99%) but is never identified. The failure is terrain-general.

Site	Terrain	MA@100m	MA@200m	MA@500m	Median err
01	riverside-urban	1.0%	1.0%	2.0%	3424 m
02	coastal-agri.	1.1%	2.2%	2.2%	3313 m
03	hilly-vegetated	0.0%	0.0%	3.0%	3304 m
08	built-up	0.0%	0.0%	0.0%	4440 m
11	arid-town	1.0%	1.0%	3.1%	3528 m
All sites		$\leq 1.1\%$	$\leq 2.2\%$	$\leq 3.1\%$	3.3–4.4 km

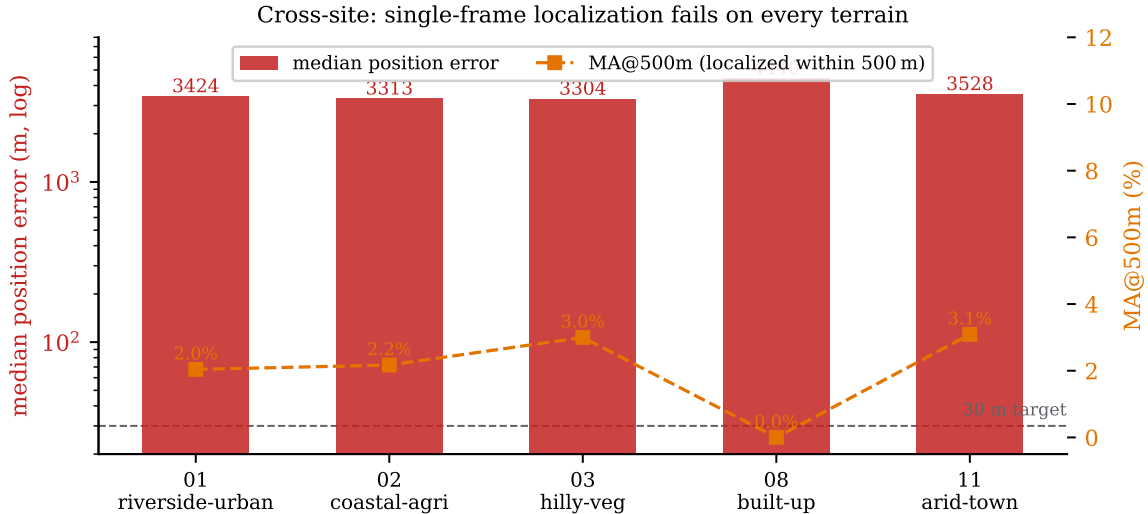


Figure 3: Cross-site single-frame localization on five terrains. Median position error (red bars, log scale) stays at kilometre scale on every terrain, and meter-level accuracy within 500 m (orange, right axis) never exceeds 3.1%—the matching failure is terrain-general. Building coverage is high (93–99% on the diagnostically analysed sites), so this is a matching failure, not a coverage one (Section 6.4).

6.4 Coverage is not the bottleneck

We first rule out the obvious failure mode. For 92.9% of queries, the true UAV location has at least one segmented satellite building within 150 m (Table 6, first row): the correct place is present in the map and is segmented. Any localization failure is therefore a *matching* failure, not a coverage or segmentation-recall failure.

6.5 The discriminability ceiling

Table 6 reports the truth-anchored retrieval and position error under a sequence of increasingly informative descriptors, all on the same segmentation output of the Changjiang-20 site ($N \approx 4.8 \times 10^5$ building triangles, 98 queries with ground truth). The finding is stark and consistent: *the geometrically correct match is retrieved in 0% of the top-20*, and the position error stays at the kilometre scale, regardless of how the descriptor is enriched. Adding building size, a log-polar building-constellation context (a Belongie-style shape context [3] computed over neighbouring building centroids), matching at the building rather than triangle level, and per-dimension standardization each *reduce the true-match descriptor distance*—by up to $16\times$ (from $\ell_1 = 38$ to 2.4 under standardization)—yet none lifts retrieval off 0% (Figure 4).

This is the key diagnosis: the failure is not that the correct match is far in descriptor space, but that the descriptor is too *low-entropy* to be *unique*. At 10^5 – 10^6 candidate elements, thousands of geometrically near-identical footprints sit as close to the query as the true one does; shrinking the true-match distance shrinks the impostors’ distances equally. Enriching per-element features cannot escape this, because the information that makes a *location* unique is not carried by any single element—it is the spatial arrangement of many, which per-element retrieval does not exploit. Only when the search is constrained to a small area (a coarse prior) does the error fall—to 146 m at $\approx 0.2 \text{ km}^2$ —but this merely reflects the window mechanically bounding the error rather than genuine identification: sub-30 m recall is 2.1%, barely above the $\approx 1.4\%$ expected from a uniform guess inside the window, and the error grows monotonically with the search area, from 146 m at 0.2 km^2 to 2.2 km at 50 km^2 (Figure 5). The descriptor contributes almost no localization signal beyond what the prior supplies.

Table 6: Discriminability analysis on Changjiang-20 ($N \approx 4.8 \times 10^5$ building elements, 98 queries). Retrieval = fraction of query triangles whose geometrically correct satellite match is in the top-20 (top-100 for the building rows); Error = median translation-consensus position error. Every descriptor enrichment shrinks the true-match distance yet leaves retrieval at 0%: the ceiling is discriminability, not distance.

Descriptor / matching variant	True-match $\ell_1 \downarrow$	Retrieval @top- $K \uparrow$	Median err (m) $\downarrow$
<i>Coverage: 92.9% of true locations have a nearby mapped building</i>			
5-D angles (baseline)	38.0	0.0%	3492
+ boundary canonicalization	18.1	0.0%	3084
+ per-dim standardization	2.4	0.0%	3543
+ building size (log A , P)	4.3	0.0%	3096
+ constellation context	—	0.0%	2102
Building-level shape	—	0.6%	3330
+ constellation context	—	0.3%	2562
5-D + context, search $\leq 0.2 \text{ km}^2$ (coarse prior)	—	1.5%	146

6.6 Sequential integration does not recover it

Since every accurate map-matching method integrates ambiguous per-frame observations over a trajectory (CFBVM’s Gaussian-mixture matches in a particle filter [28]; SWA-PF’s 4-DoF particle filter [34]), we test whether the same estimator rescues our descriptor. On a 100-frame consecutive flight segment we run a particle filter (2×10^4 particles, random-walk motion of $\approx 266 \text{ px/frame}$ matching the measured inter-frame displacement) whose per-frame observation is the Gaussian-mixture of our descriptor’s Hough votes. It does not converge (Table 7, Figure 6): the trajectory-level error stays at kilometre scale (filtered median 5.5 km, *worse* than the single-frame median of 3.2 km, because the filter commits to a spuriously consistent wrong track) and sub-100 m recall is 0%. The diagnostic explains why: the true position receives *any* vote in only 12% of frames (median 0 votes/frame; Fig-

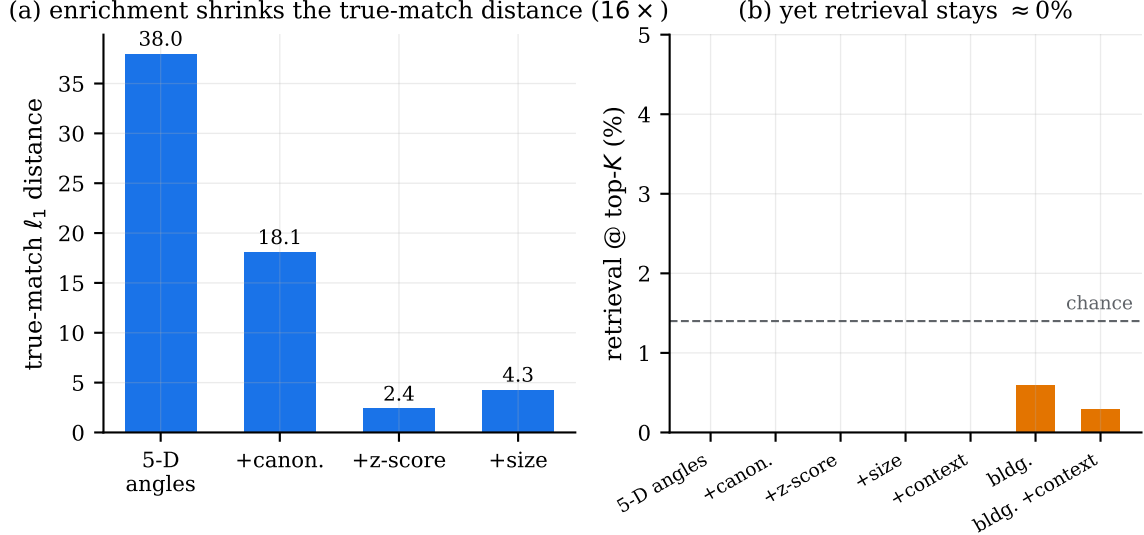


Figure 4: The discriminability ceiling on Changjiang-20 (4.8×10^5 elements, 98 queries). (a) Successive descriptor enrichments shrink the distance between a query triangle and its geometrically correct satellite match by up to $16\times$ (ℓ_1 38 $\rightarrow$ 2.4). (b) Yet the fraction of queries for which that correct match is retrieved among the top- K candidates stays at essentially 0% across every variant—below the chance rate expected inside the search window. Shrinking the true-match distance shrinks the impostors’ distances equally; the descriptor is too low-entropy to be unique at scale.

ure 6a), and when present its region ranks ≈ 123 rd. A particle filter integrates a signal that *recurs*; ours is absent from 88% of frames, so the filter locks onto a spurious track (Figure 6b). The limiting factor is thus the *observation model*: unlike CFBVM’s area-density descriptor, our per-element geometric descriptor does not provide recurring, spatially consistent support for the true location, which no downstream filter can manufacture. Figure 7 makes this concrete on one frame—the descriptor’s votes scatter across the entire map rather than concentrating at the true position.

Table 7: Two-phase pipeline on a 100-frame Changjiang-20 trajectory. Phase 2 (particle-filter integration of the per-frame Phase 1 matches) does not recover localization, because Phase 1’s observation supports the true position in only 12% of frames.

Estimator	Median error (m)↓
Phase 1 only (single-frame Hough vote)	3214
Phase 1 + Phase 2 (particle filter)	5507
Frames with any support at true position	12%
Sub-100 m recall	0%

Runtime and scale. The pipeline is nonetheless efficient and compact, as intended: descriptor extraction and a single K -NN query take ≈ 0.8 s per image on one CPU core, and the site-01 index of 4.8×10^5 elements occupies ≈ 12 MB ($5 \times \text{float32}$ per element). These costs scale linearly in the number of indexed elements; the barrier to city-scale use is accuracy, as analyzed above, not compute or memory.

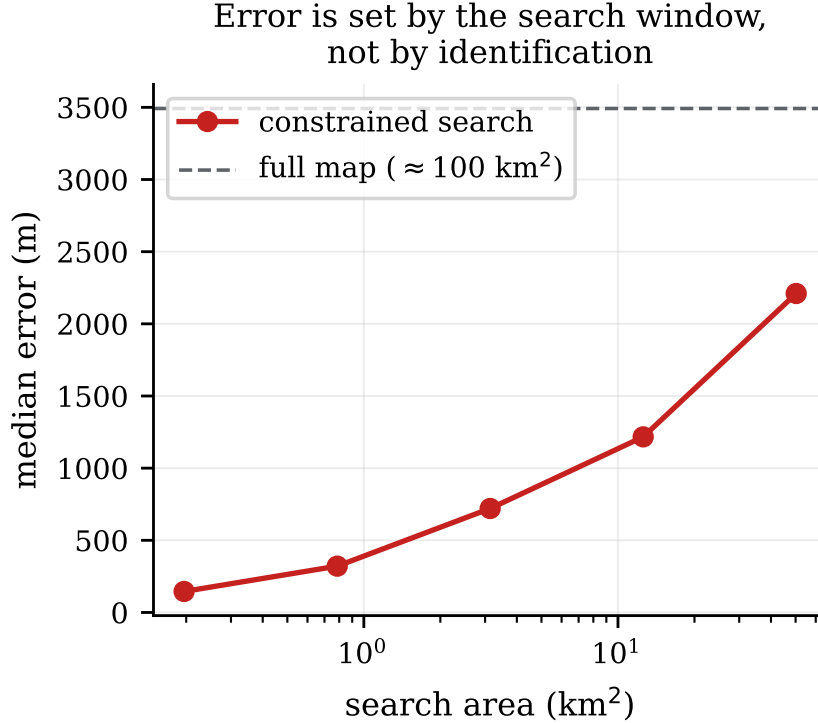


Figure 5: Localization error versus the area of the (coarse-prior) search window. The median error grows monotonically with the search area—from 146 m at 0.2 km² to 2.2 km at 50 km², approaching the full-map error (dashed)—showing that the accuracy in a small window is set by the window bounding the error, not by the descriptor identifying the location.

6.7 Qualitative Visualization

Figure 8 shows Phase 1’s front end on a representative building-dense UAV frame: Mask R-CNN building segmentation (orange outlines) and the interior-filtered CDT mesh (blue) from which the per-triangle descriptors are computed. Segmentation and triangulation are visibly reliable; the failure analyzed in Section 6.5 arises downstream in matching, not here. The matching-stage counterpart is Figure 7 (Section 6.6): the descriptor votes for a query frame scatter across the entire map rather than concentrating at the true position.

6.8 Discussion of the failure

The analysis above identifies a single dominant failure mode and rules out several plausible alternatives. It is *not* primarily a coverage problem (the true location is mapped in 93% of cases), nor a segmentation-density problem specific to sparse terrain (the ceiling appears on dense-urban Changjiang), nor a voting or refinement problem (the correct match is absent from the top-20 before any voting), nor an estimator problem (a particle filter cannot converge on it). It is a *discriminability* problem intrinsic to per-element geometric matching: individual building triangles—even enriched with size and neighbourhood context—are not unique among 10^5 – 10^6 candidates, and the discriminative information that identifies a location lives in the joint spatial configuration of many buildings, which per-element retrieval discards. The one regime in which the descriptor localizes is when an external coarse prior shrinks the candidate set (Section 6.5), which is precisely how successful vector-map methods such as CFBVM [28] deploy their matching stage.

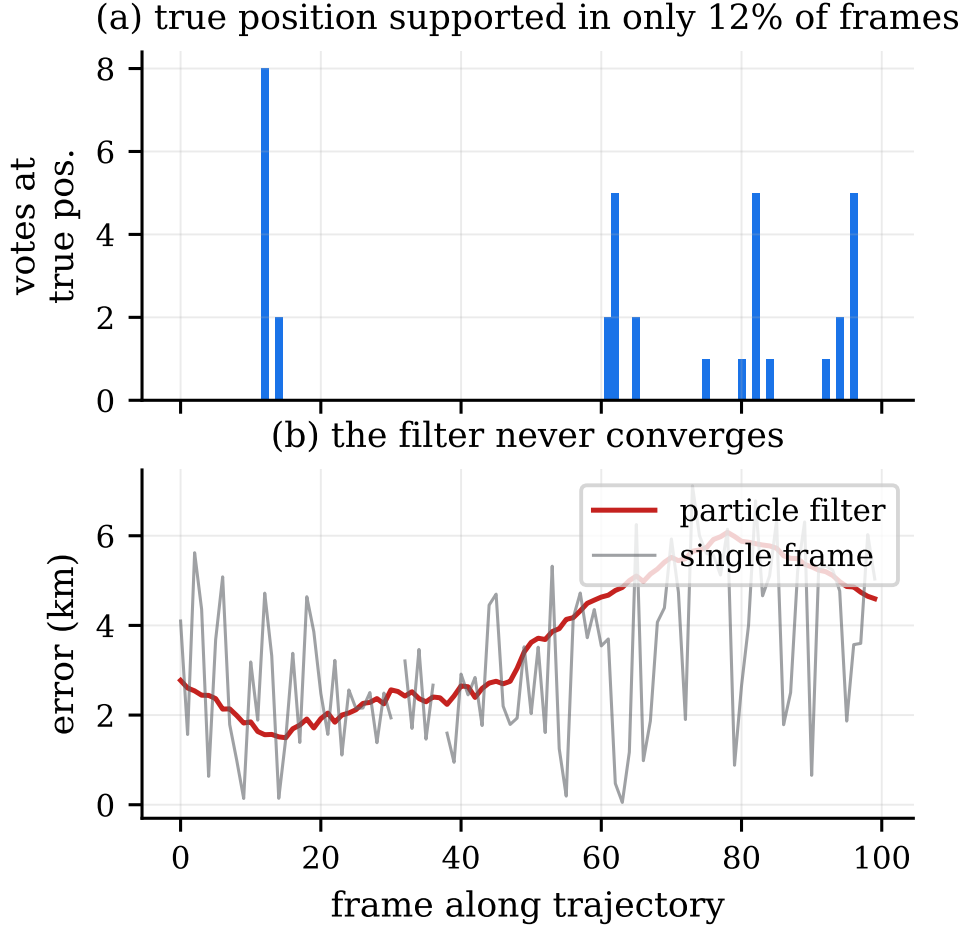


Figure 6: Why sequential integration fails. (a) Votes landing at the true position, per frame along the trajectory: the true location is supported in only 12% of frames (median 0). (b) Position error per frame: the particle filter (red) never converges and drifts onto a wrong track, no better than the raw single-frame estimate (grey). A filter cannot integrate a signal that is absent from 88% of frames.

6.9 Area-level aggregation recovers discriminability

The diagnosis above is also constructive: if the discriminative signal lives in the joint configuration of many buildings rather than in any single element, then the descriptor should be aggregated over an *area* rather than matched per element. We test this directly while keeping the Ekeland angle central. We tile each map into $\approx 200\text{m}$ cells and describe every cell by a translation-invariant descriptor: building density, layout moments (centroid spread, mean nearest-neighbour spacing), a footprint-size histogram, and—the shape signal a density field lacks—*histograms of the per-building Ekeland free-cone angles*. Each UAV frame is matched, as a single cell, against the satellite cells; we report coarse-localization recall (fraction of frames whose true cell is in the top-50 of $\sim 10^3$ cells) and compare a *plain* descriptor (density + layout + size, the information a CFBVM-style area-density field carries) against the same descriptor *augmented with the Ekeland-angle histograms*.

Two findings follow (Table 8, Figure 9). First, area-level matching *escapes the per-element ceiling*: coarse recall rises from the flat 0% of single-element retrieval to several times chance (chance $\approx 3.7\%$ at top-50). Second, and central to this paper, the *Ekeland-angle histograms add discriminability that pure density and layout do not*: on all four analysed sites they raise top-50 recall over the plain descriptor—from 5.5% to 9.1% (urban), 2.0% to 10.2% (coastal), 3.2% to 6.4% (hilly), and 0% to

(frame 01_0013.JPG, best-supported)



Figure 7: Descriptor votes (orange) for a single, best-supported query frame, overlaid on the Changjiang-20 map; the true UAV position is the green star. The votes scatter across the entire $\approx 100 \text{ km}^2$ map instead of concentrating at the truth—a direct visualization of the discriminability ceiling.

1.7% (arid)—and reduce the median coarse-localization error on three of the four. The improvement is unanimous in direction across distinct terrains, which is what makes it credible despite modest absolute magnitudes. The free-cone angle thus supplies exactly the per-building shape information that arrangement- or density-only descriptors (the CFBVM regime) discard—but at the area level, where per-element segmentation noise is averaged out rather than fatal.

We are deliberately precise about scope. These are *coarse*-localization gains: absolute recall remains single-digit-to-low-tens of percent, and the top-1 cell is still usually wrong, so the method does not yet deliver precise metric localization. The contribution is a measured, robust demonstration that (i) the area is the right matching unit for this problem and (ii) the Ekeland free-cone angle is a discriminative, complementary shape cue at that level. Closing the remaining gap to precise localization plausibly requires a learned area embedding or a clean vector map (as used by shape-and-relationship methods on ‘.shp’ e-maps [31]); both are natural extensions of the Ekeland area descriptor introduced here.

7 Conclusion

We have presented a scene-agnostic UAV visual geo-localization framework whose core novelty is a purely geometric 5-D triangle descriptor combining sorted interior angles and edge-anchored Ekeland Free-Cone Angles (MFCA). The key design innovation is the *kernel expansion* P_{sub} : the maximal polygon formed by merging CDT triangles while keeping all three original triangle vertices on the boundary. This stopping criterion is geometrically analogous to Ekeland’s cone-criticality condition, and the resulting Ekeland angle is computed exactly in $O(1)$ per vertex via an interior-angle closed form, replacing the earlier 1° -sweep approximation. The descriptor is invariant to texture, colour, resolution, and season, and requires no scene-specific training. The design of the MFCA component is

Segmentation + CDT mesh (01_0001.JPG)



Figure 8: Phase-1 front end on a Changjiang-20 query frame: Mask R-CNN building footprints (orange) and the interior-filtered CDT mesh (blue) that yields the 5-D per-triangle descriptors. Segmentation and triangulation are reliable; matching is the bottleneck (Section 6.5).

Table 8: Area-level matching (≈ 200 m cells, $\sim 10^3$ candidates, 100 frames/site). Coarse recall @top-50 and median coarse error for a plain density+layout+size descriptor (CFBVM-style) vs. the same descriptor augmented with Ekeland free-cone-angle histograms. The Ekeland histograms improve recall on every site (chance $\approx 3.7\%$).

Site	Terrain	Recall@top-50	Median err
<i>Plain (density + layout + size; CFBVM-style)</i>			
01	urban	5.5%	3467 m
02	coastal	2.0%	5136 m
03	hilly	3.2%	4760 m
11	arid	0.0%	4379 m
<i>+ Ekeland free-cone-angle histograms (ours)</i>			
01	urban	9.1%	4311 m
02	coastal	10.2%	4080 m
03	hilly	6.4%	3205 m
11	arid	1.7%	3530 m

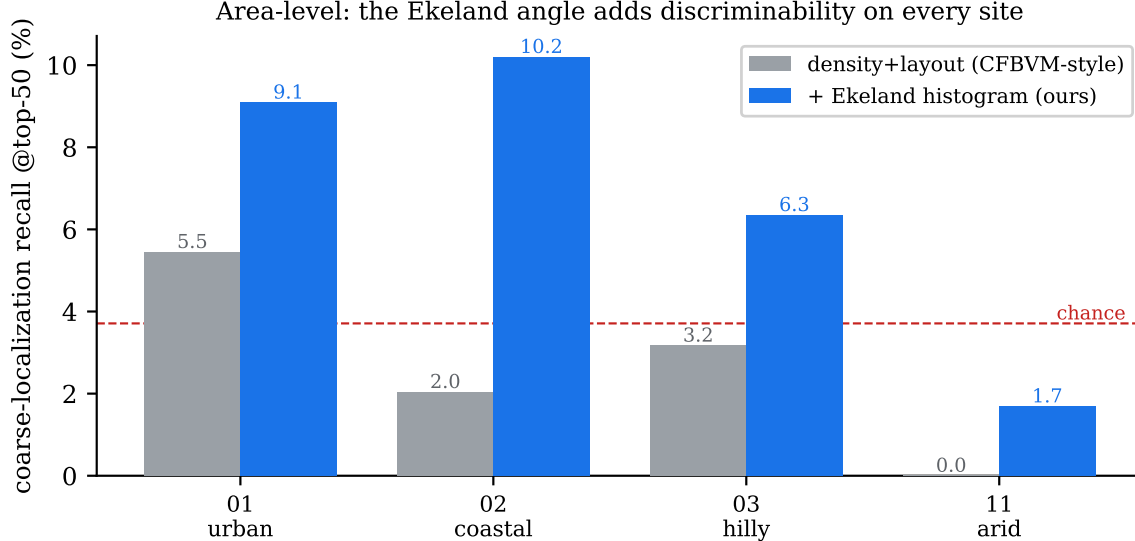


Figure 9: Area-level coarse-localization recall (@top-50) per site: a plain density+layout descriptor (CFBVM-style, grey) vs. the same descriptor with Ekeland free-cone-angle histograms added (ours, blue). The Ekeland angle lifts recall on every terrain (dashed line: chance). The gain is unanimous in direction, isolating the free-cone angle as a complementary shape cue that arrangement/density descriptors lack.

geometrically inspired by Ekeland’s variational principle [12]: each element of the principle (complete metric space, bounded function, cone-touching condition, critical point value) has a natural geometric counterpart in our construction (Table 2)—an analogy that motivates the descriptor’s design and yields a closed-form computation with no sup-not-attained edge case, rather than a literal instantiation of EVP’s analytic machinery. An offline KD-tree index reduces online matching to $O(M \log N)$; plurality voting over satellite patches converts the matching output to a predicted 2-D geographic position.

Our second contribution is an honest, quantitative characterization of this descriptor as a UAV-localization matching stage. On UAV-VisLoc we find that, although building coverage is high (93%), single-frame per-element retrieval never places the correct match in the top-20 and localization error stays at the kilometre scale; that enriching the descriptor with size, a building- constellation shape context, building-level granularity, and dimension normalization shrinks the true-match distance up to $16\times$ without moving retrieval off zero; and that integrating the per-frame matches over a trajectory with a particle filter—the mechanism that makes CFBVM and SWA-PF succeed—also fails to converge, because the true position receives recurring observation support in only 12% of frames. The limiting factor is therefore the *observation model*: purely geometric per-element matching does not, on its own, provide the recurring, spatially consistent support that scale-robust localization requires. This is a useful negative result. It shows that the promising direction is not a richer per-element descriptor but a fundamentally more discriminative *observation*—an area-level or configuration-level descriptor (as in CFBVM’s density rings) or a learned embedding—paired with the coarse-to-fine, sequential estimation that the accurate methods already employ. Our released code and analysis pipeline (`eval.py` and the diagnostic scripts) let future work measure any candidate observation model against the same retrieval, coverage, and sequential-support criteria.

Our third contribution acts on this diagnosis and, crucially, keeps the Ekeland angle central (Section 6.9). Aggregating the descriptor over ≈ 200 m cells—as a histogram of per-building Ekeland free-cone angles alongside building density and layout—escapes the per-element discriminability ceiling: coarse-localization recall rises from a flat 0% to several times chance. More importantly, across all four analysed sites the Ekeland-angle histograms improve coarse recall over a density/layout-only (CFBVM-style) descriptor—for example, top-50 recall $2.0\% \rightarrow 10.2\%$ on the coastal site—and reduce

median error on three of four. This isolates the Ekeland free-cone angle as a discriminative shape cue at the area level that arrangement- or density-only descriptors discard. The gains are coarse in absolute terms, but unanimous in direction across distinct terrains, and they point to the concrete next step: a learned area embedding, or matching against clean vector maps [31], built on the Ekeland area descriptor introduced here.

Limitations and scope. (i) Our multi-site evaluation covers five of the eleven UAV-VisLoc sites, selected for terrain diversity (urban, coastal, hilly, built-up, arid); the discriminability ceiling was consistent across all five and the area-level Ekeland gain was unanimous in direction, but the remaining six sites and the full per-site error distributions are left to future work. (ii) Our area-level result is a coarse-localization study (top- K cell recall among $\sim 10^3$ cells); we do not yet demonstrate precise metric localization, which we expect requires either a learned area embedding or a clean vector map, and did not implement. (iii) The method assumes approximately nadir views, buildings in the field of view, and heading to $\pm 5^\circ$; open terrain without buildings is out of scope. (iv) SWA-PF [34] is, as of this writing, an arXiv preprint with no peer-reviewed venue found; its numbers are reproduced as reported and should be re-checked against a peer-reviewed version if one appears.

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